B2B

Environmental analysis of the B2B scheme, testing solar exposure, prevailing winds and seasonal daylight to inform the design of a garden dome housing a cafe, growing beds, processing and research spaces.



Solar, wind and daylight studies were carried out across the winter, equinox and summer solstices, testing the dome's orientation, glazing and internal zoning to ensure daylight and shelter for the growing, processing and gathering spaces year-round.

01/09/2019 - date started

02/04/2020 - date ended

Documents by Sebastian Dawber
https://www.sebastianvictordawber.co

Created for ...
eg BSc Architecuture and enviromental design
Undergraduate RIBA ARB part I

All work produced by Sebastian Victor Dawber

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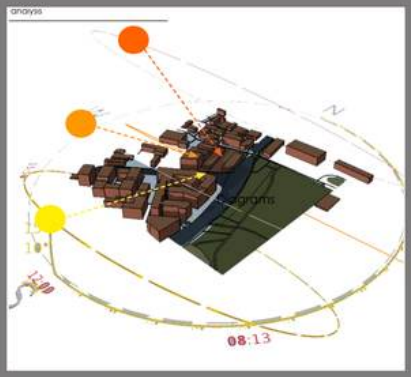
SEBASTIAN VICTOR DAWBER
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BSc Part I, ARB RIBA , Architecture and environmental design

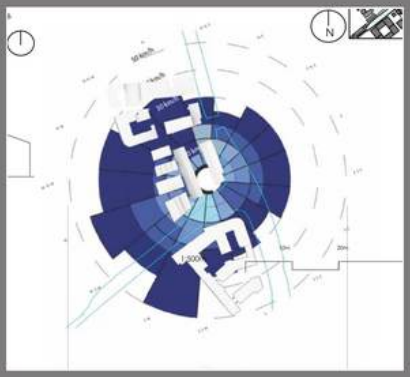
B2B

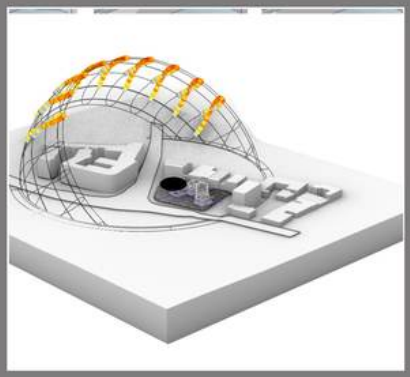
Solar, Wind & Daylight Analysis

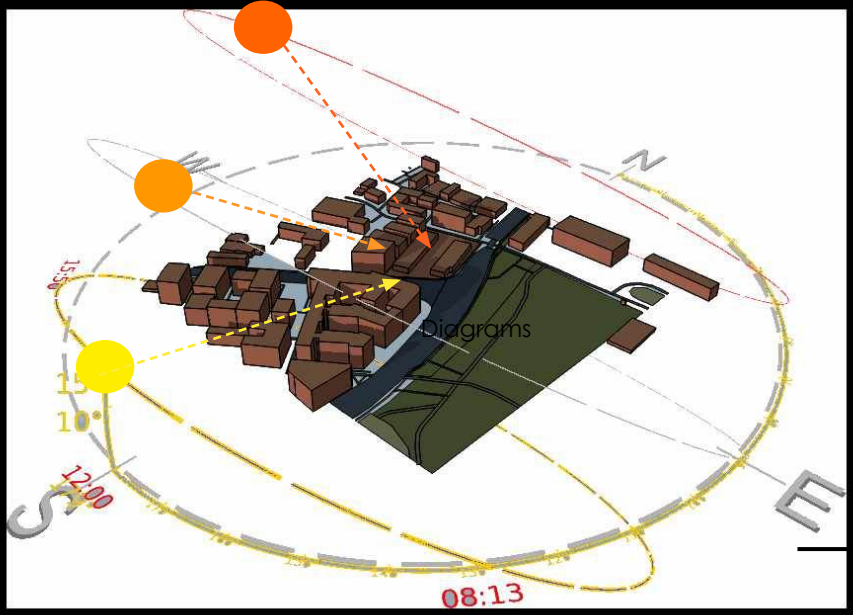
02/04/2020 - date ended

This document presents an environmental analysis carried out for the B2B project, a mixed-use scheme bringing together a cafe, garden, processing and research spaces beneath a geodesic dome. Solar radiation, prevailing wind and daylight performance were tested across the winter, equinox and summer solstices to understand how the site's climate shapes the internal environment. Wind testing identified areas of shelter and exposure across the site, while daylight testing at 1m, 3m, 6m and 9m heights measured how much light reaches the growing and gathering spaces throughout the day. Together, these studies informed decisions on the dome's orientation, glazing distribution and internal zoning, ensuring the spaces below receive appropriate light and shelter from prevailing wind throughout the year.

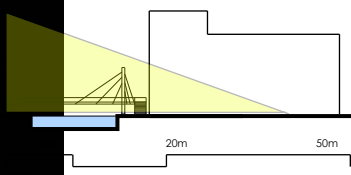






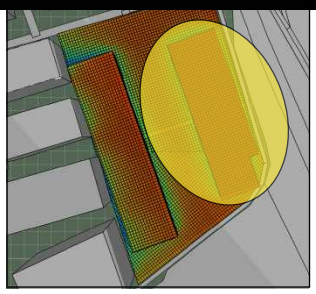


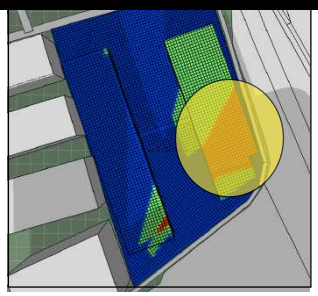


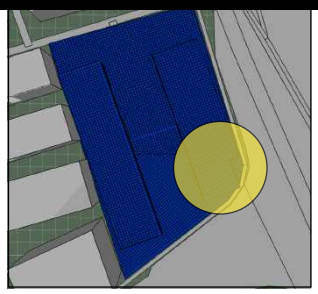


m

WINTER

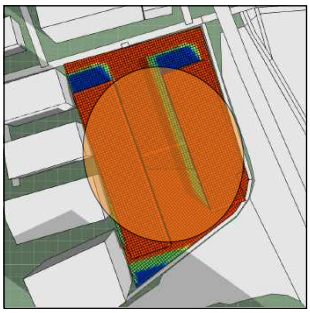


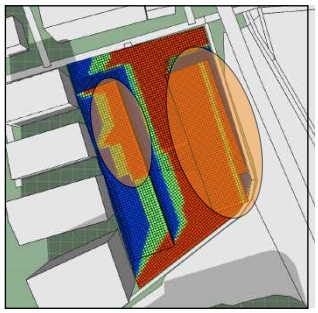


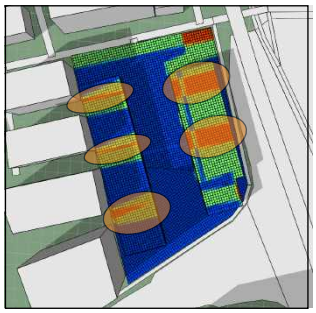


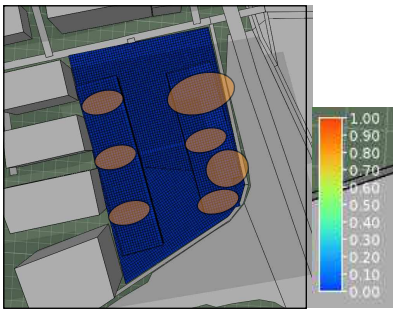
After 16 pm the sun goes down and due to the low sun angle the site does not receive light after 15pm

equinox

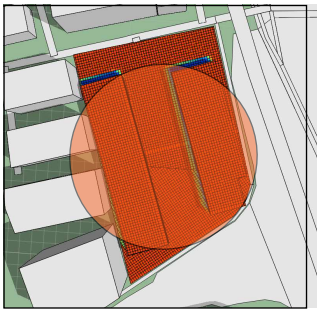


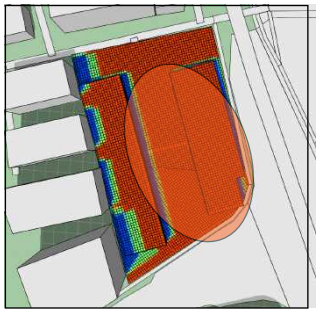


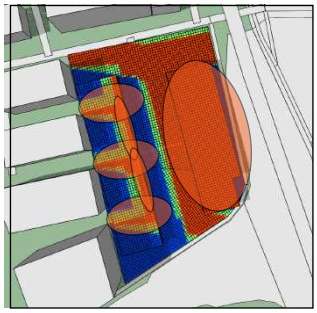


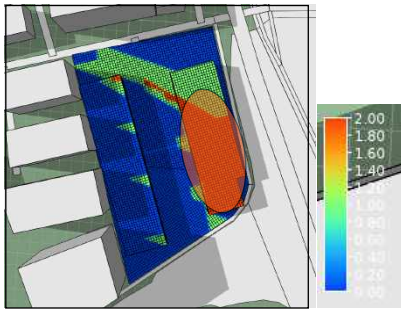


Summer









	1	15.5	15.7	15.6
	2	15.5	15.7	15.6
	3	15.5	15.4	15.5
	4	15.5	15.5	15.8
	5	15.5	15.7	15.5
	6	15.5	15.6	15.4
	7	15.8	15.6	15.5
	8	16	15	15.4
	9	15.7	15.6	15.3
	10	15.8	15.4	15.3
	11	15.9	15.6	15.4
	12	15.4	15.7	15.6
	13	15.6	15.6	

Winter solstice
- sun spots (marked)
- The low angle of the sun and the adjacents buildings create shadows and block the sun effecting most of the site

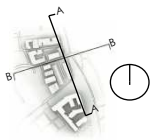
Equinox solstice
- sun spots (marked)
-After 15pm the site starts to lose sun due to the sun setting in the West the adjacent buildings in the West block out the low angle sun due to sunset at 18pm

Summer solstice
- sun spots (marked)
- In summer the sun recives suffiecent light throughout the day due to the higher angle of the sun. When it starts to set the West side buildings effect the amount of sun that reaches the site

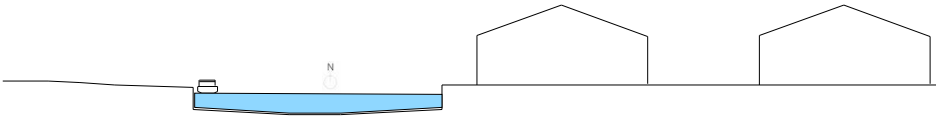
Wind analysis

Down draft effect

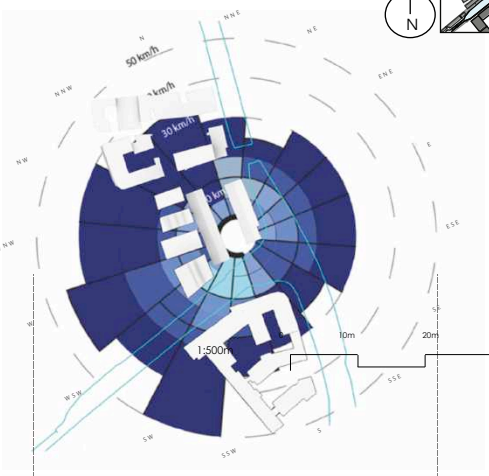


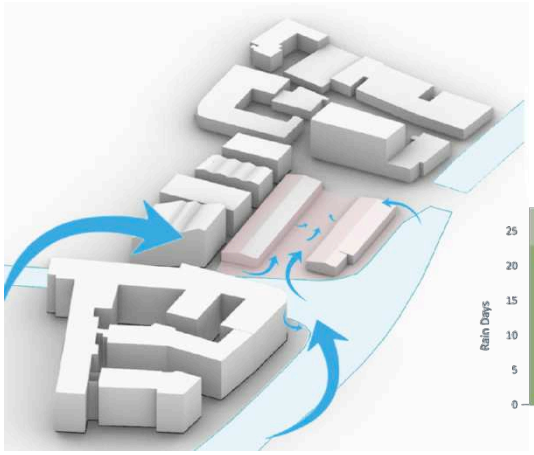


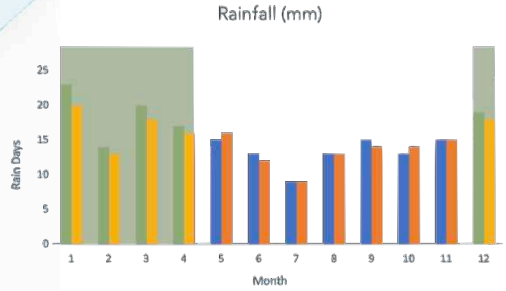
BB

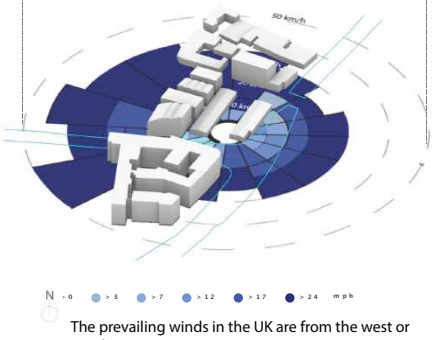


Wind Path



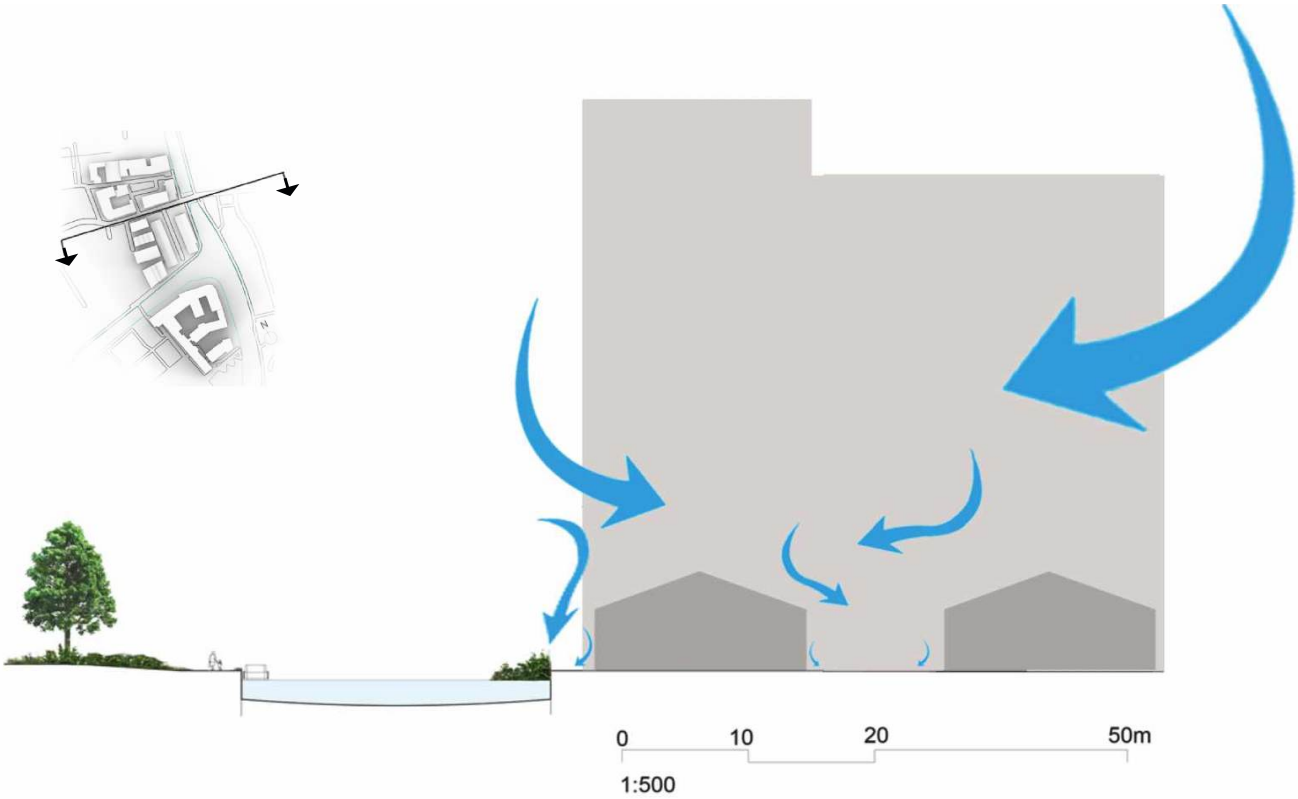




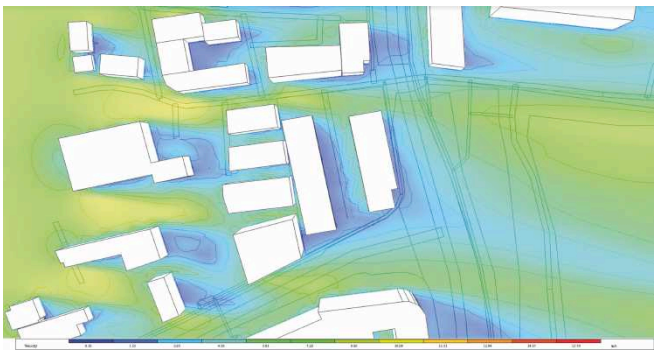


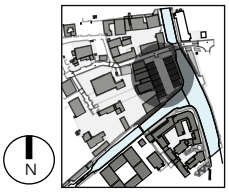
South - East Axonometric view showing direction of wind speed.

The prevailing winds in the UK are from the west or South-

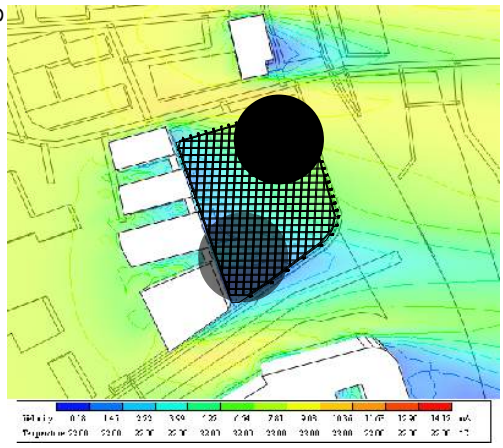


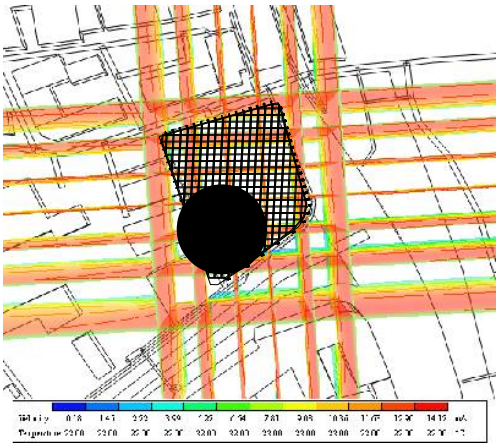
Further analysis was done of the environment using IESVE.From the analysis we can understand that the site has a wind speed between 1.5 m/s to 5 m/s depending on the location.Downdraft effect is created from the building opposite having aapprox. wind speed of 7m/s. Therefore, the heights of the buildings did affect the direction of the wind.The buildings surrounding the site created a barrier in the center.However, from the results obtained during analysis,there wasn't strong winds on the day except for the down draft bursts of wind.

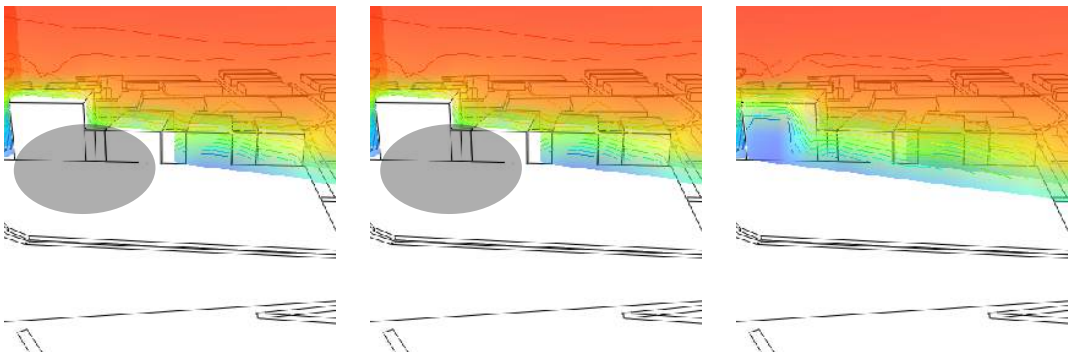




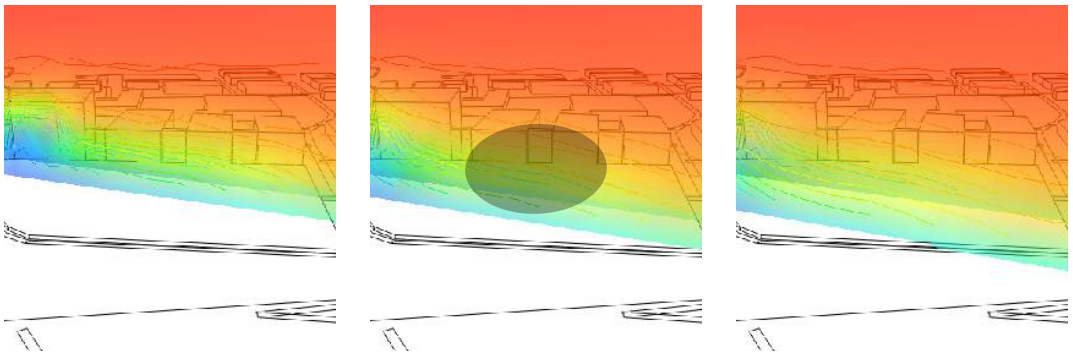
5m testing



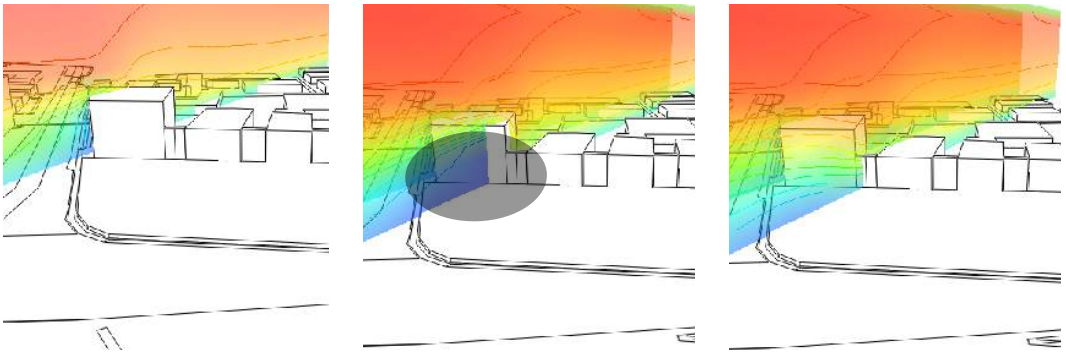




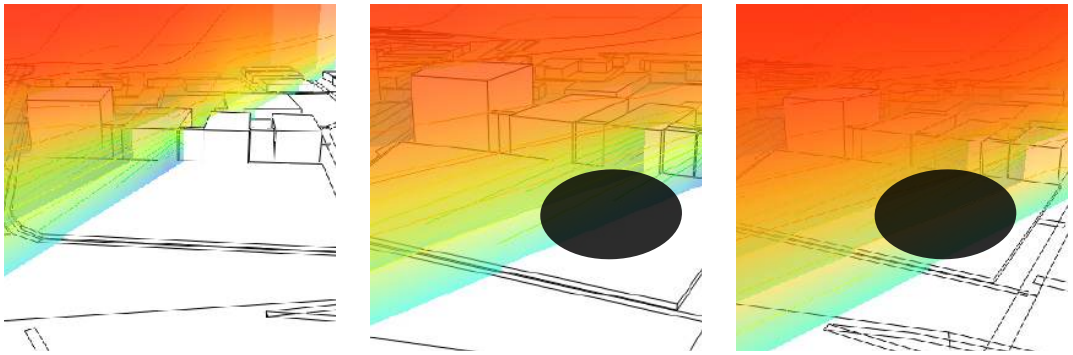
Protected area



Nutural area



Wake effect



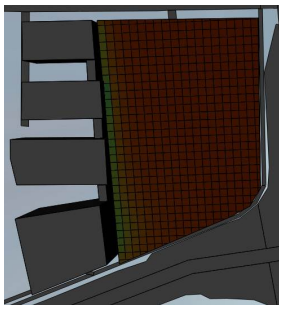
Stronger winds

22nd December
WInter solstice

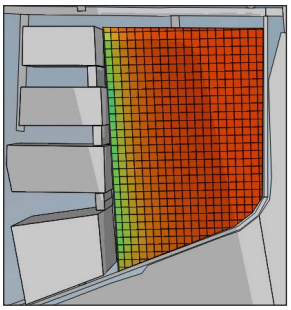
Testing for cafe, garden, processing and research spaces

1m

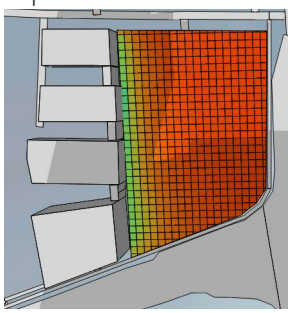
8am



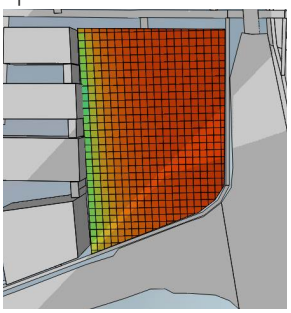
10am



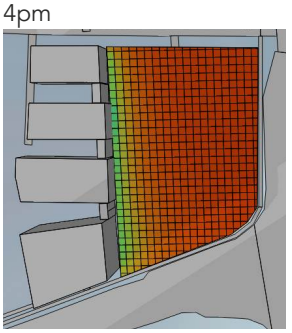
12pm



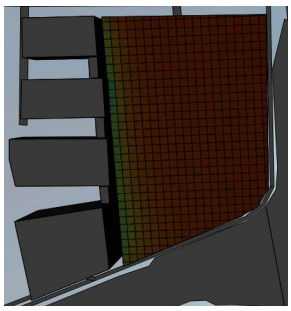
2pm

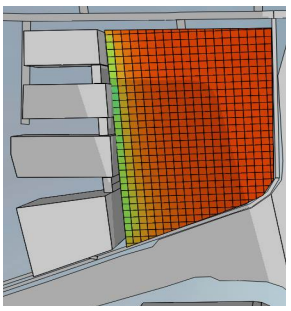


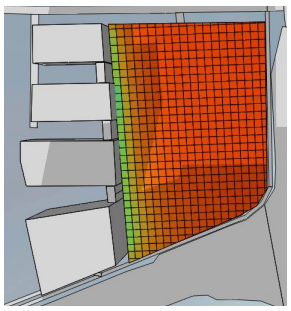
4pm

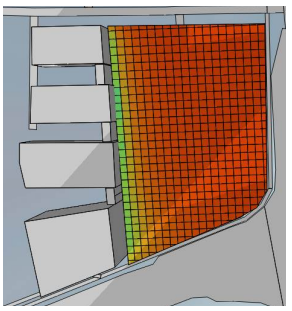


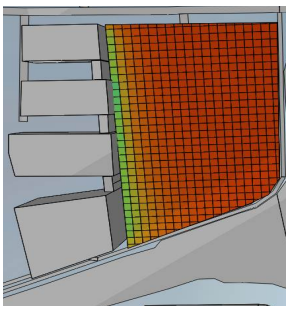
3m



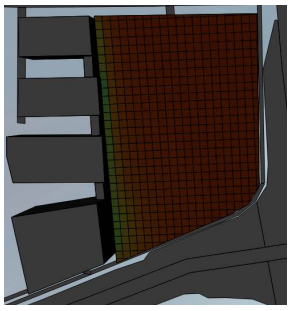


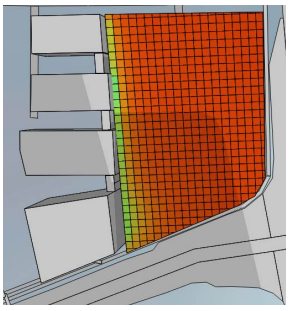


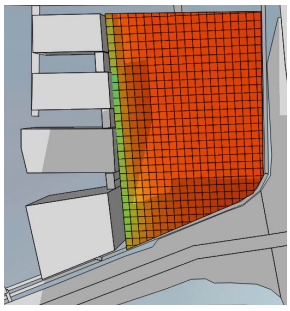


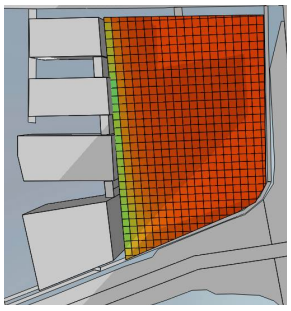


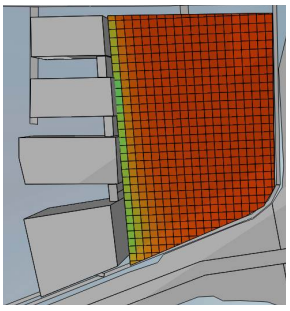
6 m



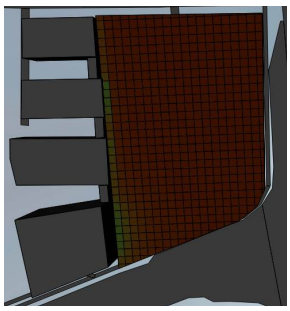


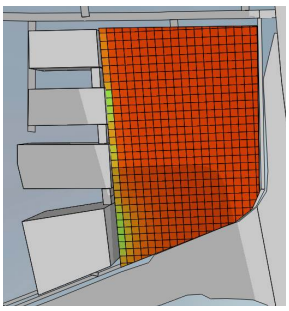


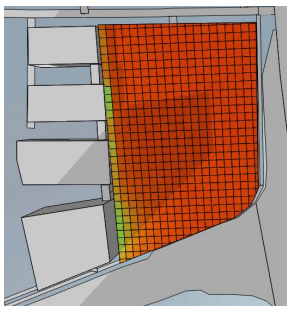


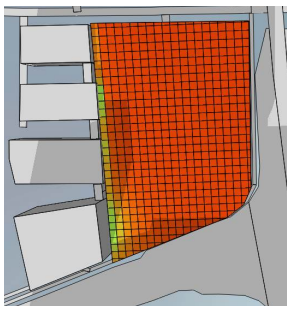


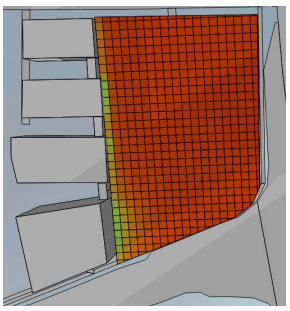
9 m

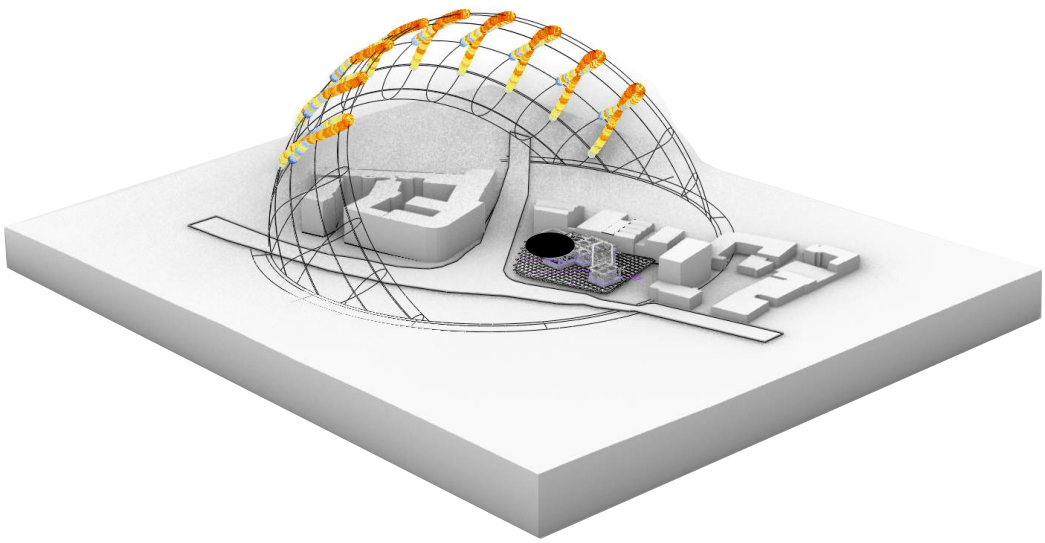










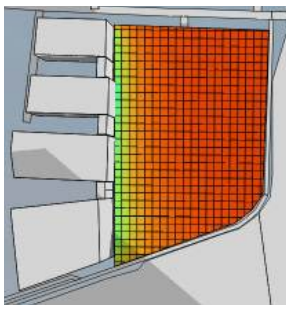


22nd March
Equinox solstice

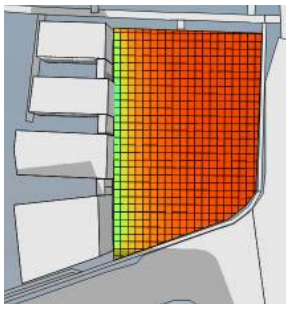
Testing for cafe, garden, processing and research spaces

1m

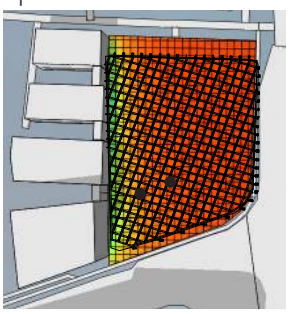
8am



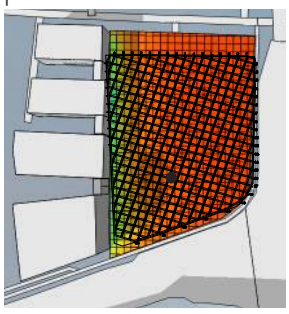
10am



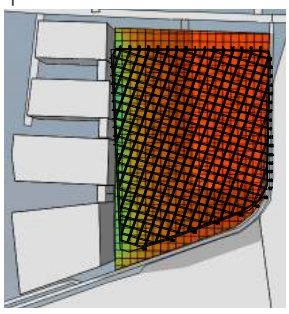
12pm



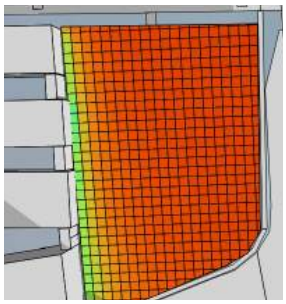
2pm

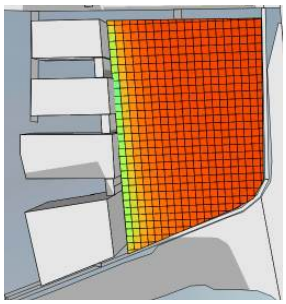


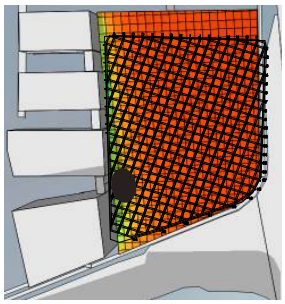
4pm

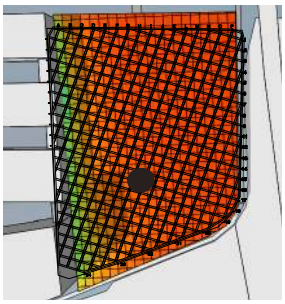


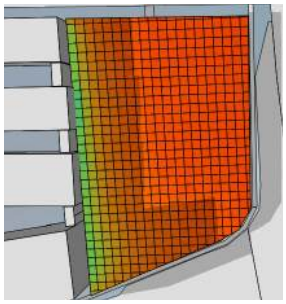
3m



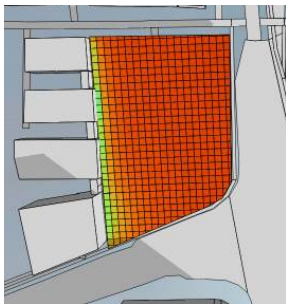


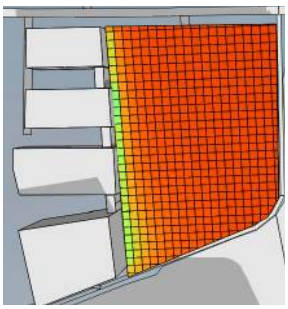


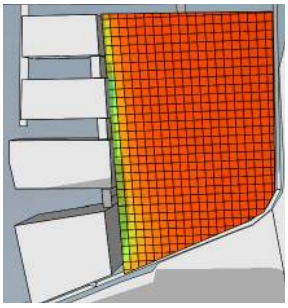


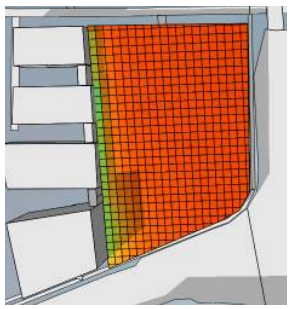


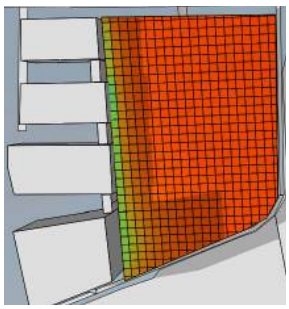
6 m



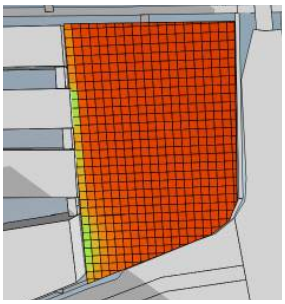


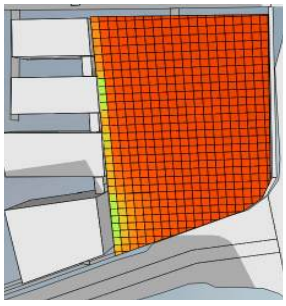


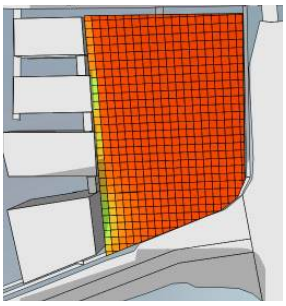




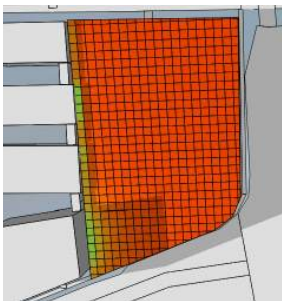
9 m

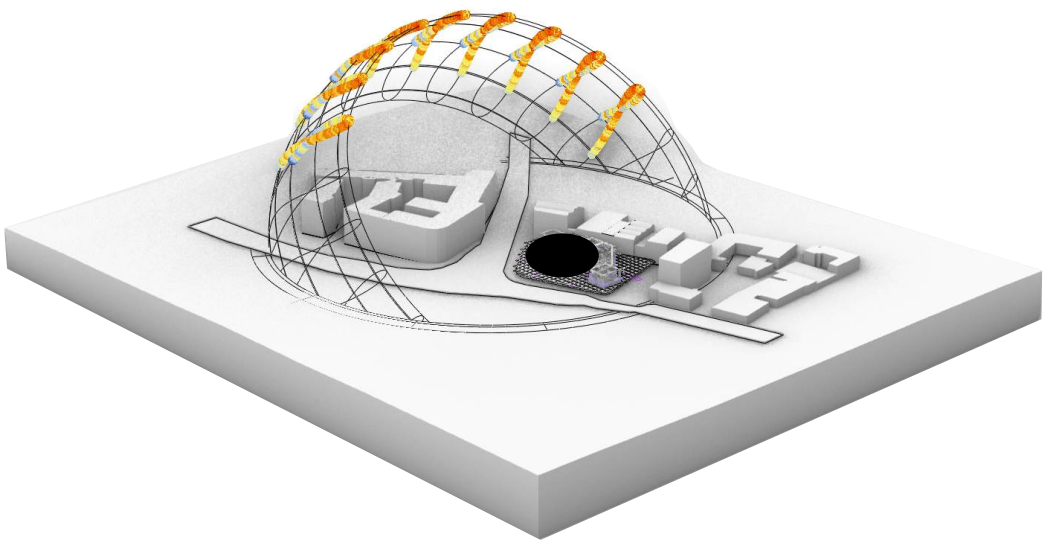










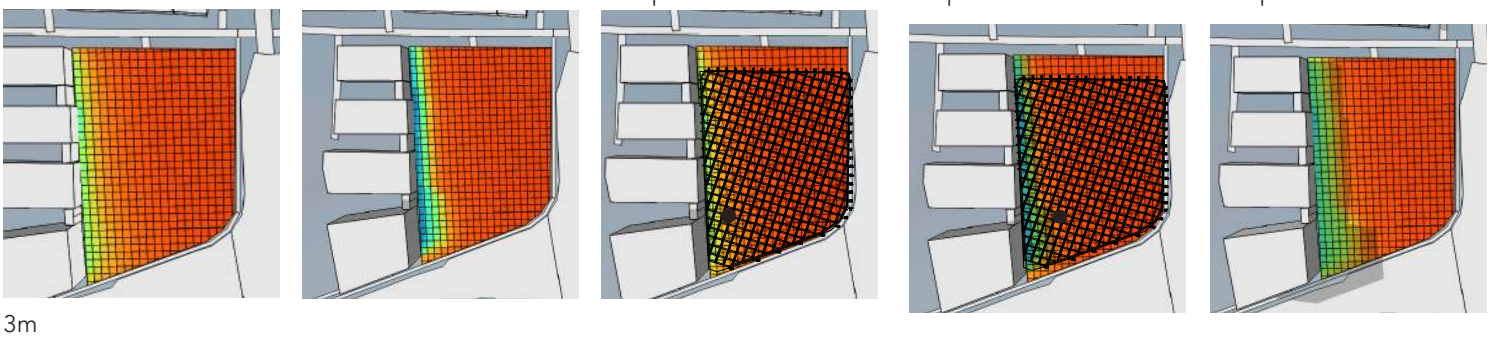


22nd June
Summer solstice

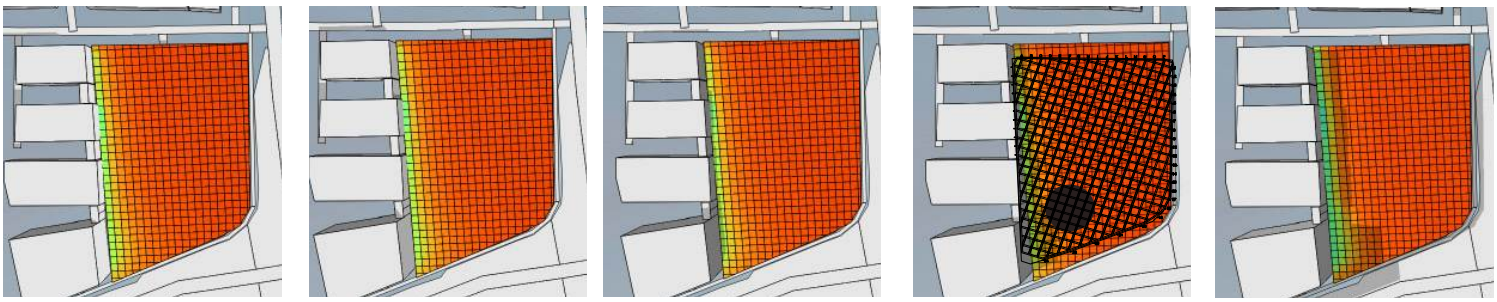
Testing for cafe, garden, processing and research spaces

1m

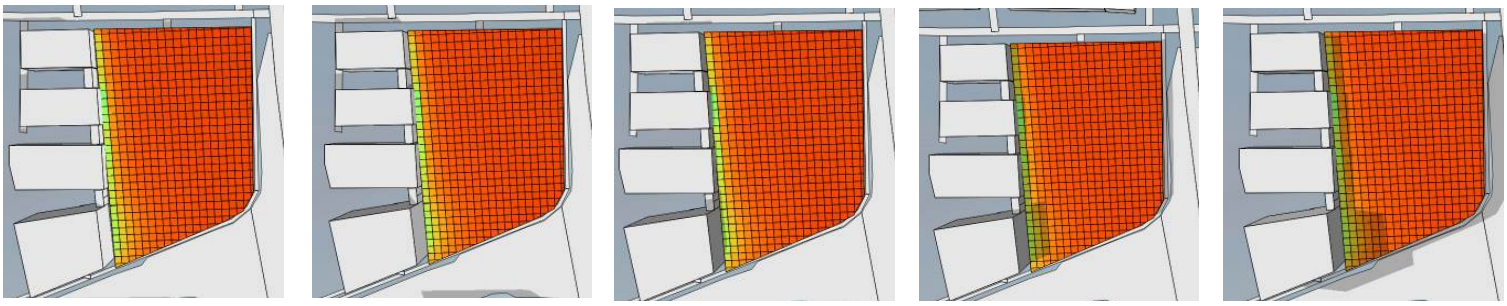
8am 10am 12pm 2pm 4pm



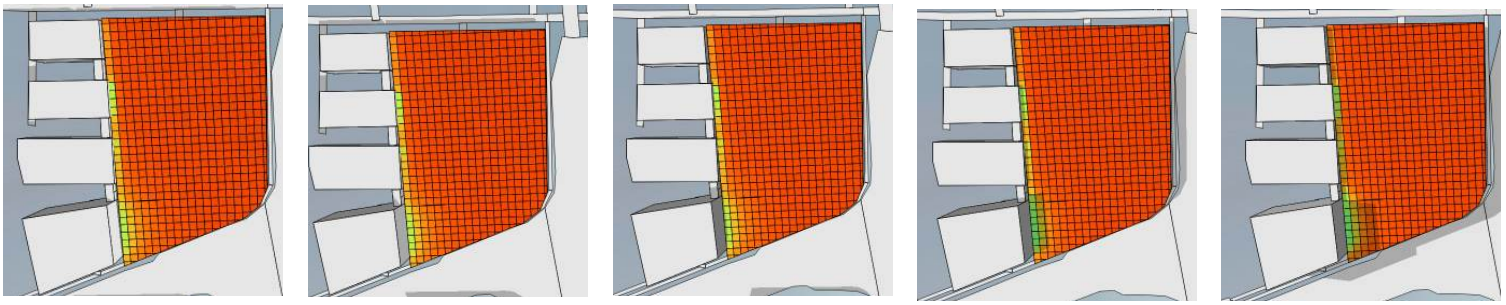
3m

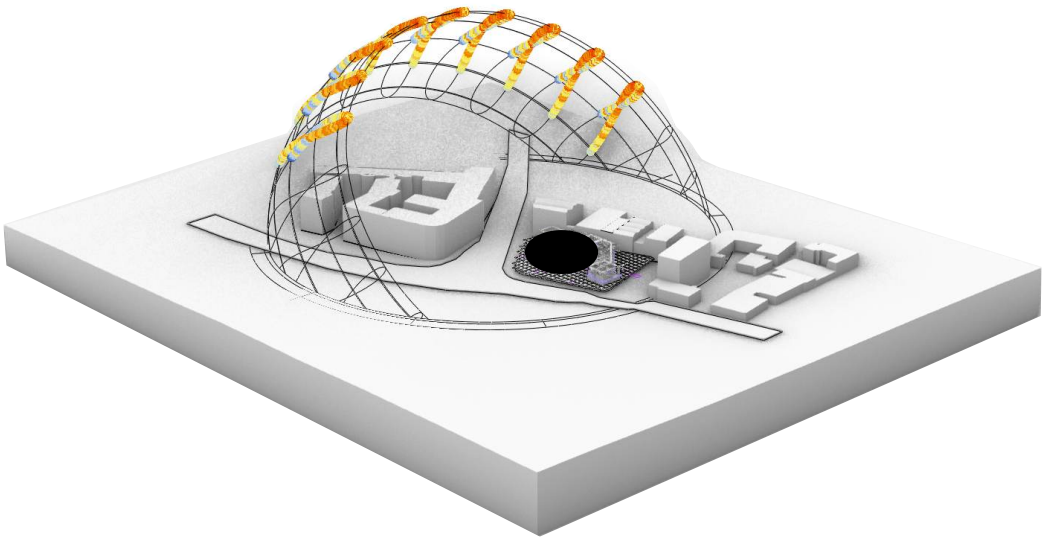


6 m



9 m





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Research into ..



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BSc Architecture and environmental design
Undergraduate RIBA ARB part I

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2019 – 2020

This document summarises the environmental analysis carried out for the B2B project, covering solar, wind and daylight testing across the seasons. The findings shaped the geometry, glazing and internal zoning of the garden dome, and can be used as a reference for the environmental strategy of future mixed-use or bioclimatic developments combining growing, processing and community spaces.



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